Year 13 Further Maths - Powers and roots of complex numbers

Similar questions - Roots of unity and polynomials

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 You are given $\omega = \cos\frac{2\pi}{7} + \mathrm{i}\sin\frac{2\pi}{7}$ and the equation $z^7 = 1$.

(a) Show that ω is a root of the equation and write down the other six roots in terms of ω. **[3]**

(b) Show that $1 + \omega + \omega^2 + \ldots + \omega^6 = 0$. **[2]**

(c) Let $t = \omega + \dfrac{1}{\omega}$. Show that $t^3 + t^2 - 2t - 1 = 0$. **[4]**

(d) Hence show that $2\cos\frac{2\pi}{7}$ is a root of $t^3 + t^2 - 2t - 1 = 0$. **[1]**

2 The fourth roots of unity are represented on an Argand diagram by the points A, B, C, D, and P, Q, R, S are the midpoints of the sides AB, BC, CD, DA of the square $ABCD$.

(a) Write down the fourth roots of unity and the complex numbers represented by P, Q, R, S. **[3]**

(b) Determine a polynomial equation of degree 8 with integer coefficients whose roots are the complex numbers represented by A, B, C, D, P, Q, R, S. **[5]**

Question 1

(a) $\omega^7 = \cos 2\pi + \mathrm{i}\sin 2\pi = 1$ (de Moivre). The other roots are $\omega^2, \omega^3, \omega^4, \omega^5, \omega^6$ (and 1)

(b) The roots of $z^7 - 1 = 0$ sum to 0 (coefficient of z^6), so $1 + \omega + \ldots + \omega^6 = 0$

(c) Divide by ω^3: $\omega^{-3} + \omega^{-2} + \omega^{-1} + 1 + \omega + \omega^2 + \omega^3 = 0$, i.e. $\left(\omega^3 + \omega^{-3}\right) + \left(\omega^2 + \omega^{-2}\right) + \left(\omega + \omega^{-1}\right) + 1 = 0$. Now $\omega^2 + \omega^{-2} = t^2 - 2$ and $\omega^3 + \omega^{-3} = t^3 - 3t$. So $t^3 - 3t + t^2 - 2 + t + 1 = 0$, i.e. $t^3 + t^2 - 2t - 1 = 0$

(d) $t = \omega + \omega^{-1} = 2\cos\frac{2\pi}{7}$ (since ω^{-1} is the conjugate), so it is a root

Question 2

(a) $A = 1, B = \mathrm{i}, C = -1, D = -\mathrm{i}$. $P = \frac{1+\mathrm{i}}{2}$, $Q = \frac{-1+\mathrm{i}}{2}$, $R = \frac{-1-\mathrm{i}}{2}$, $S = \frac{1-\mathrm{i}}{2}$

(b) The fourth roots of unity satisfy $z^4 = 1$. For P: $\left(\frac{1+\mathrm{i}}{2}\right)^2 = \frac{2\mathrm{i}}{4} = \frac{\mathrm{i}}{2}$, so $P^4 = -\frac{1}{4}$; similarly for Q, R, S. So $z^4 = -\frac{1}{4}$, i.e. $4z^4 + 1 = 0$. Polynomial: $(z^4 - 1)(4z^4 + 1) = 4z^8 - 3z^4 - 1 = 0$